JINAL VANIYA

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Summary

Data and AI Analyst with ~2 years of experience building LLM-powered automation, reporting, and data validation systems across healthcare, telecom, aerospace, and academic research. Cut 4K viewership reporting time from 24+ hours to under 5 mins, reduced invalid telemetry records by 25% at ISRO, and analyzed 900+ LLM responses for demographic bias. Skilled in Python, SQL, Power BI, RAG, and LangChain.

Education

Hofstra University **Jan 2025 - May 2027 (Expected)**
Master of Science in Data Science (GPA: 3.92)

Vishwakarma Government Engineering College **July 2015 - June 2019**
Bachelor of Engineering (B.E.) in Information Technology

Work Experience

AI Analyst Intern | Hotwire Communications Ltd. | Florida, USA **April 2026 - Present**

- Developed and automated a 4K viewership reporting workflow in Databricks using Genie and a custom AI agent, auto-generating reports for 20+ games with ~98% accuracy (validated through team testing with senior analysts) and cutting turnaround from 24+ hours to under 5 minutes.
- Reviewed 12 AI-driven Wi-Fi automation use cases against compliance requirements, using Claude and VS Code to resolve rule inconsistencies, diagnose automation failures, and improve system reliability.
- Maintain and enhance "Ask Toby," an internal HR RAG chatbot built on OpenWebUI and ChromaDB, testing 15+ queries to identify and report 5 response-accuracy issues and assisting in testing PII filters that protect sensitive employee data.

Research Assistant | Hofstra University | New York, USA **Oct 2025 - March 2026**

- Designed and conducted a research study evaluating demographic bias in LLM-generated toy advertisements, engineering N-shot prompting frameworks to generate 100+ toy scripts and score model responses on a 1-5 scale across 25+ product categories.
- Applied NLP techniques (NLTK, part-of-speech tagging, set operations) to 900+ responses to quantify linguistic and demographic biases, supported by statistical analysis and interpretation of findings.
- Built LangChain-based prompting pipelines integrating LLM APIs for document processing and intelligent querying, persisting 100+ structured outputs in JSONL and deploying automated Python workflows on AWS EC2 for scalable, reproducible analysis.

Data Analyst | Indian Space Research Organization (ISRO) | Gujarat, India **Feb 2021 - Feb 2022**

- Analyzed and validated satellite telemetry streams at ~100 Mbps, running integrity checks across data channels to maintain data accuracy, reducing invalid records by 25%. Engineered a Flask-based web interface that centralized and simplified system access, reducing user onboarding time and standardizing the training workflow.

Projects

Knee Osteoarthritis Severity Grading | Python, PyTorch, Swin Transformer, ResNet, Streamlit (GitHub)

- Developed a multimodal deep learning system utilizing 9,786 knee X-rays with clinical metadata to classify osteoarthritis severity across five categories. Benchmarked nine model configurations, achieving a test macro F1-score of 0.8312, an 18.5% relative improvement over the strongest image-only model using multimodal fusion and ordinal classification.
- Built a Streamlit application supporting image-only and multimodal inference, returning predicted severity grades, confidence scores, and educational summaries.

Clinical RAG Chatbot | Python, LangChain, Groq, FAISS, Streamlit (GitHub)

- Developed a RAG chatbot that answers clinical questions from a 750+ page medical reference, combining FAISS vector search over page-level chunks with low-latency Groq-hosted LLM inference.
- Implemented source attribution returning three cited pages per answer for user verification, and validated accuracy and relevance through manual testing on medical queries.

Hospital Emergency Room Dashboard | Power BI, Power Query, Excel (GitHub)

- Designed an interactive Power BI dashboard analyzing 9,200+ emergency room records to monitor patient volume, average wait time, patient satisfaction, and department referrals. Engineered the dataset in Power Query, deriving age-group and admission-hour fields and classifying each visit against a 30-minute wait-time target. Standardized gender labels and built a calendar table to enable time-based trend analysis.

Technical skills

Programming & Databases: Python, SQL, R, PostgreSQL, MongoDB, Java
Machine Learning & AI: PyTorch, Scikit-learn, NLP (NLTK), LLMs, LangChain, RAG, FAISS, ChromaDB, Prompt Engineering (Few-Shot/N-Shot), Deep Learning (Swin Transformer, ResNet)
Data Analysis & Reporting: Pandas, NumPy, KPI Reporting, Reporting Automation, Statistical Analysis, Data Visualization
BI & Analytic Tools: Power BI (DAX, Power Query), Tableau, Excel, Databricks (Genie)
Platforms & Tools: Jupyter notebook, VS Code, AWS (EC2), Docker, Git, Flask, Streamlit, OpenWebUI

Certifications

Microsoft Azure AI Essentials Professional Certificate – Microsoft
Generative AI Professional Certificate – Snowflake
AI for Everyone - DeepLearning.AI
Stanford Machine Learning - Coursera